power

File: schematics/power.kicad_sch

fpga

File: schematics/fpga.kicad_sch

usb_jtag_uart

File: schematics/usb_jtag_uart.kicad_sch

flash_microsd

File: schematics/flash_microsd.kicad_sch

j21_interface

File: schematics/j21_interface.kicad_sch

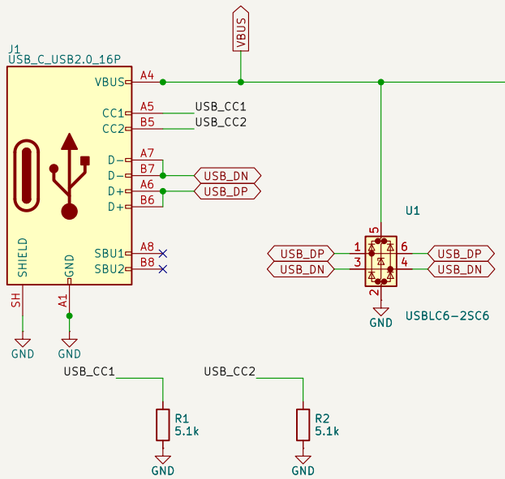
Preserved blocks: Tang Nano 9K 3674 (schematic + iBOM); FT2232H: FTDI DS v2.10 fig. 6.1
Pre-release (README rev 0.4): allocation and new parts are Proposed; open items in README section

Sheet: /
File: tangnano9k-kr260-hat.kicad_sch

Title: Tang Nano 9K Custom HAT for AMD Kria KR260

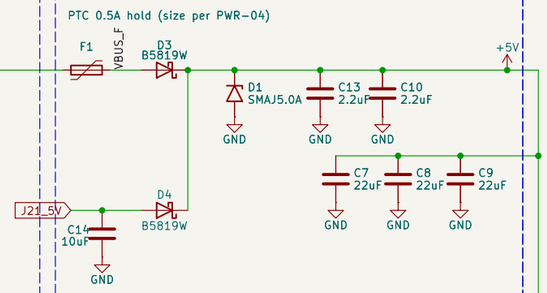
Size: A4	Date: 2026-10-21	**Rev: 0.4**
KiCad E.D.A. 10.0.7		Id: 1/6

USB-C input (power and FT2232H USB)



CC1/CC2 with 5.1k (USB-C device). D+/D- protected by the USBLC6-2SC6.

Input protection and +5V

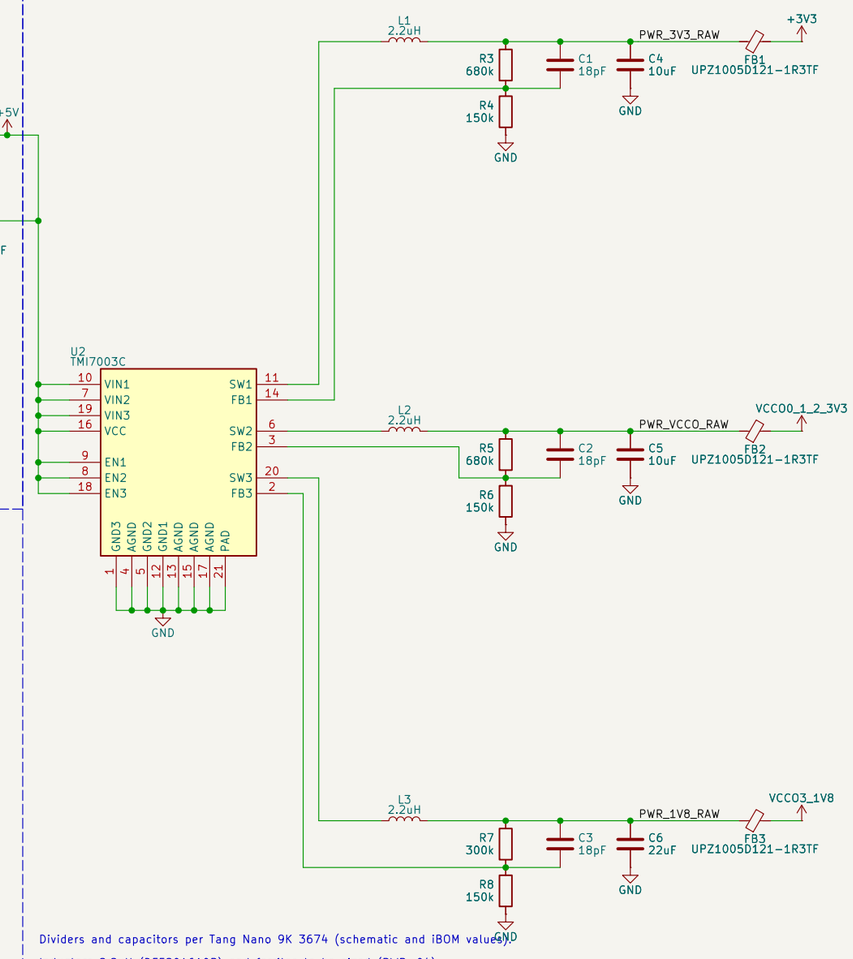


F1: PTC, current rating to be sized (PWR-04).

D1: unidirectional TVS, cathode on +5V.

D3/D4: power from USB-C or from the KR260 J21 5V; Schottky OR-ing, no back-feed between the sources.

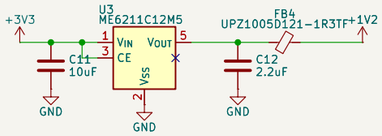
TMI7003C: triple buck (pinout from Tang Nano 9K 3674; datasheet Pending)



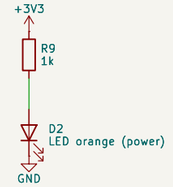
Dividers and capacitors per Tang Nano 9K 3674 (schematic and iBOM values).

Inductors 2.2uH (DFE201610P) and ferrites to be sized (PWR-04).

LDO 1V2 (ME6211C12M5)



Power LED



Rail test points and PWR_FLAG



Preserved blocks: Tang Nano 9K 3674 (schematic + iBOM); FT2232H: FTDI DS v2.10 fig. 6.1

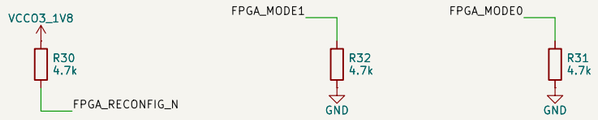
Pre-release (README rev 0.4): allocation and new parts are Proposed; open items in README section

Sheet: /power/
File: power.kicad_sch

Title: Power: USB-C, TMI7003C, ME6211

Size: A3	Date: 2026-10-21	**Rev: 0.4**
KiCad E.D.A. 10.0.7		Id: 2/6

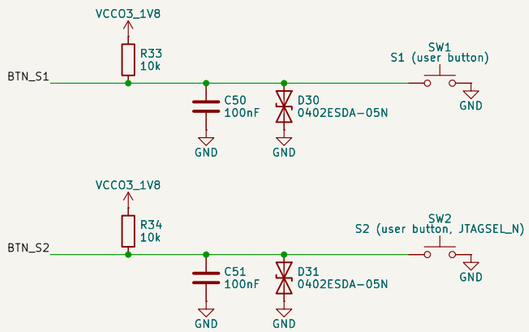
Configuration: RECONFIG_N and MODE



RECONFIG_N with pull-up and MODE0/MODE1 with 4.7k pull-downs (Tang Nano 9K 3674).

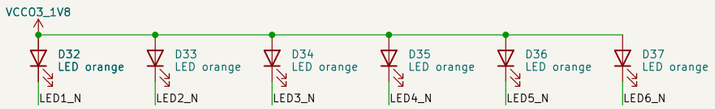
Bank 3 (VCCIO3) at 1V8: JTAG, MODE, buttons and LEDs.

Buttons S1 and S2

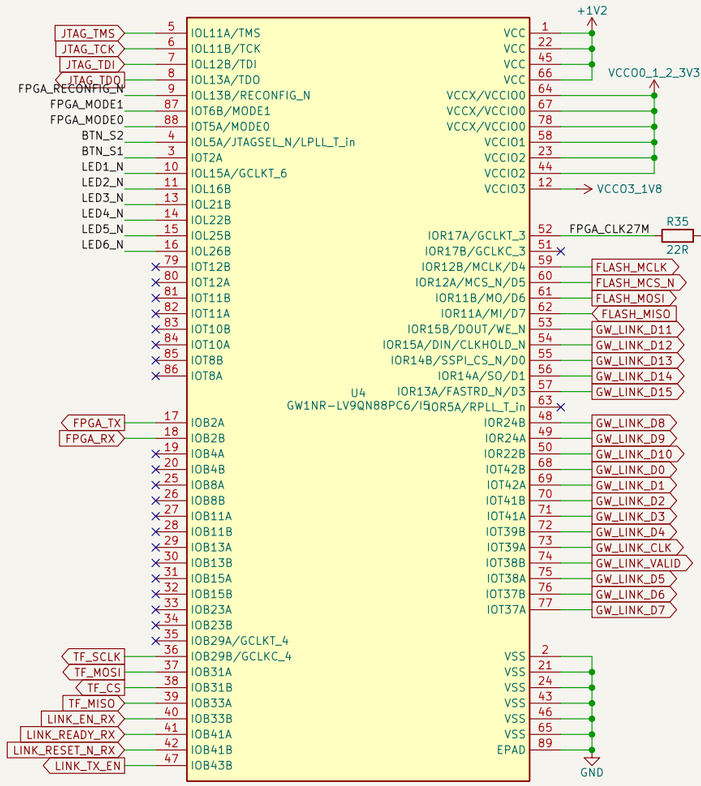


S2 is also JTAGSEL_N (pin 4). 10k pull-up, 100nF and ESD per button.

User LEDs (active low)



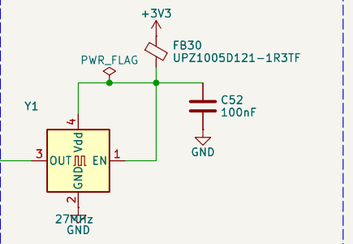
No series resistor, as on the 3674 (README 16.1).



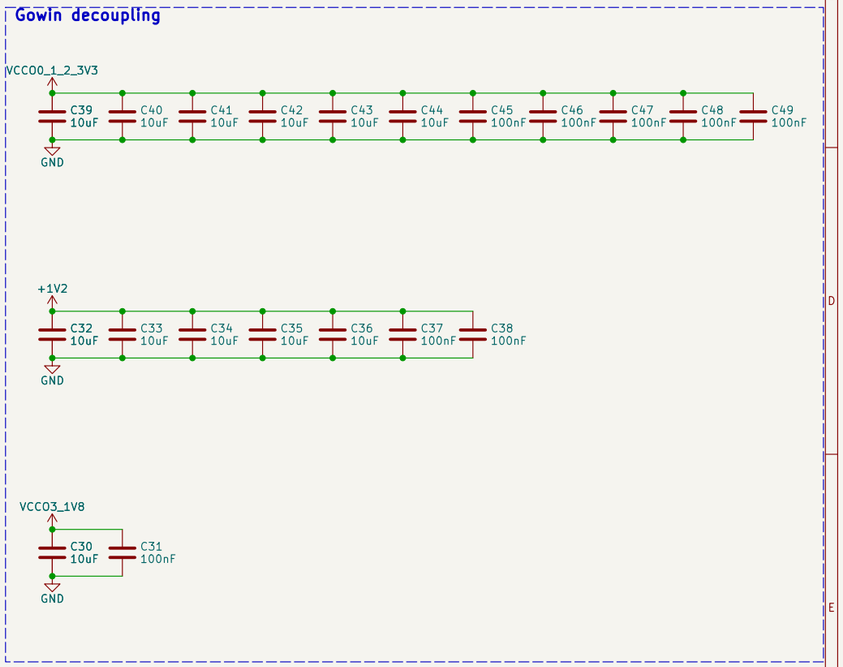
Pins 53–57 (D11–D15) also carry SSPI/CPU configuration functions: use as I/O in the open flow is Pending.

Bank 1 (pins 48–77): KR260 link; bank 2: UART, microSD and link control. Allocation: Proposed.

27 MHz oscillator



Gowin decoupling



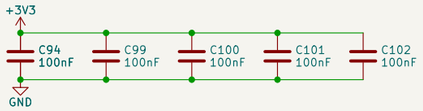
Preserved blocks: Tang Nano 9K 3674 (schematic + iBOM); FT2232H: FTDI DS v2.10 fig. 6.1
Pre-release (README rev 0.4): allocation and new parts are Proposed; open items in README section

Sheet: /fpga/
File: fpga.kicad_sch

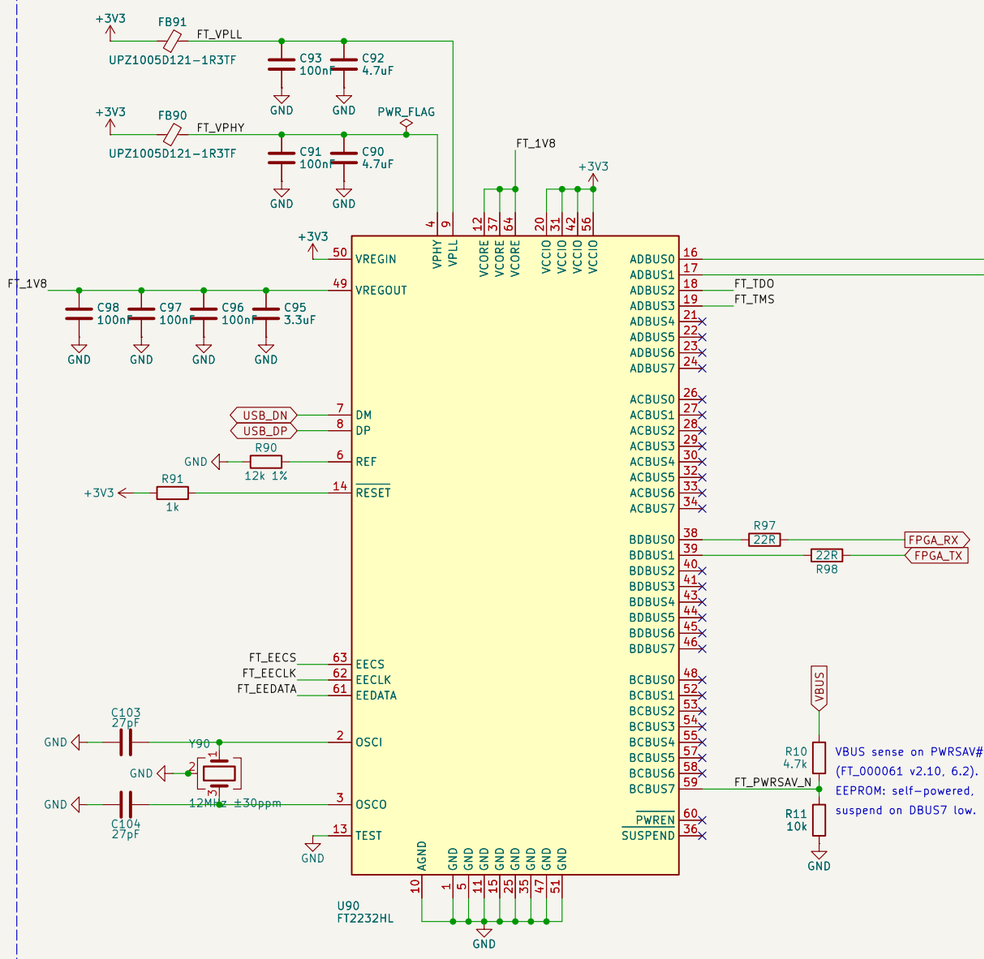
Title: GW1NR-9 QN88P, 27 MHz clock, LEDs, buttons

Size: A3	Date: 2026-10-21	**Rev: 0.4**
KiCad E.D.A. 10.0.7		Id: 3/6

+3V3 decoupling (FT2232H)

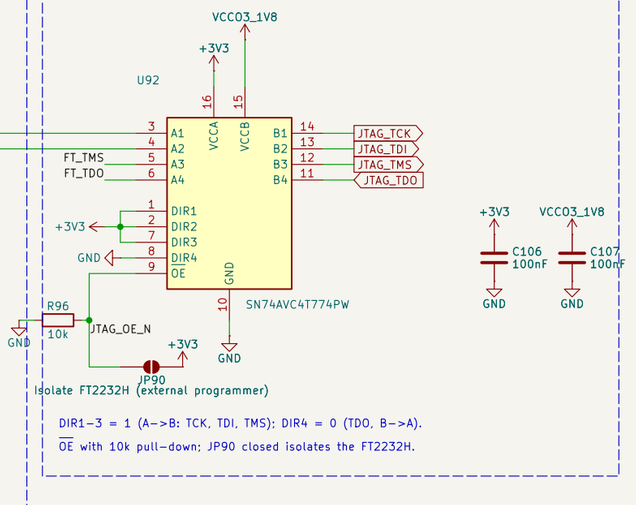


FT2232HL (FT_000061 v2.10, fig. 6.1)



Per FT_000061 v2.10 fig. 6.1. Channel A: JTAG (MPSSE); channel B: UART with 22R series.

JTAG level translator 3V3 -> 1V8 (SN74AVC4T774)

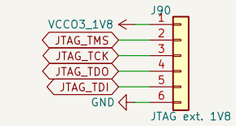


Isolate FT2232H (external programmer)

DIR1–3 = 1 (A->B: TCK, TDI, TMS); DIR4 = 0 (TDO, B->A).

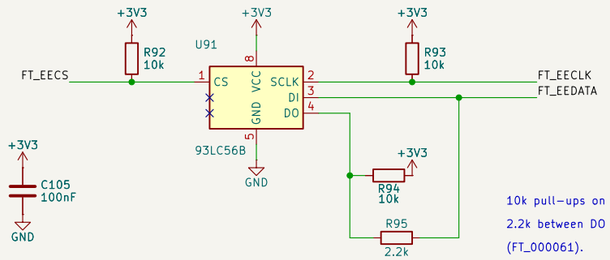
$\overline{OE}$ with 10k pull-down; JP90 closed isolates the FT2232H.

External JTAG (1V8)



Header for an external 1V8 programmer;
with JP90 closed the FT2232H is isolated.

EEPROM 93LC56B (FT2232H configuration)

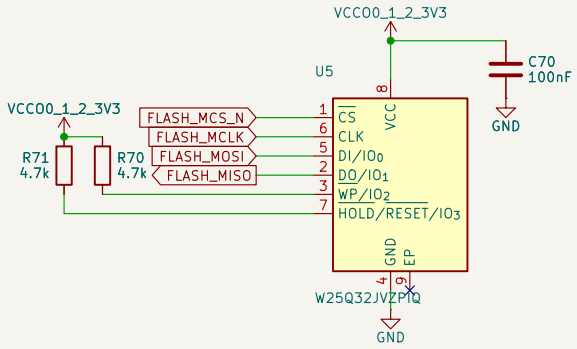


10k pull-ups on CS, CLK and DO;
2.2k between DO and DATA
(FT_000061).

Preserved blocks: Tang Nano 9K 3674 (schematic + iBOM); FT2232H: FTDI DS v2.10 fig. 6.1
Pre-release (README rev 0.4): allocation and new parts are Proposed; open items in README section

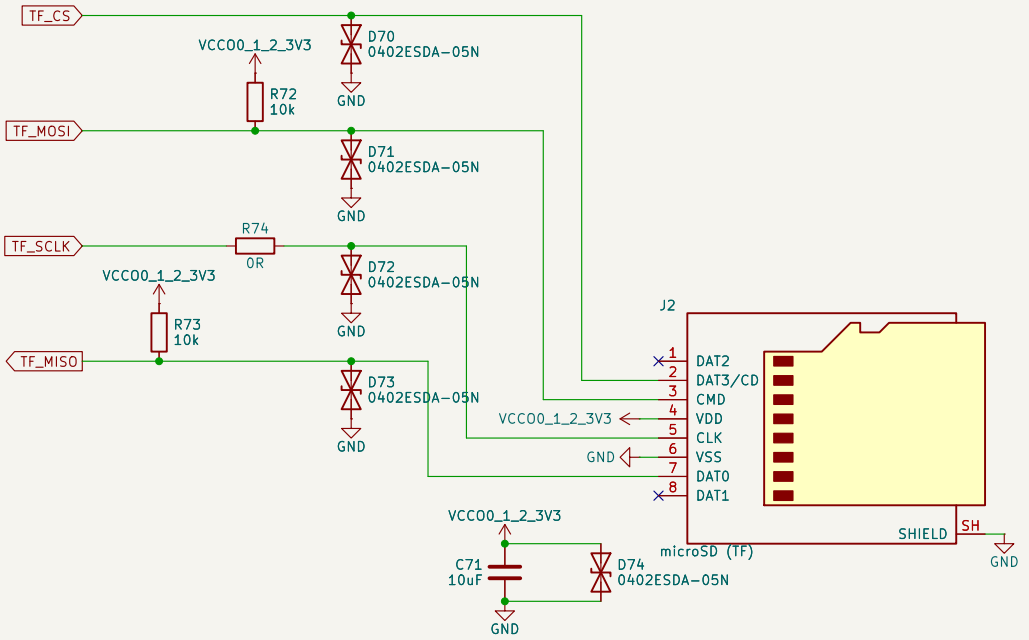
Sheet: /usb_jtag_uart/		
File: usb_jtag_uart.kicad_sch		
Title: FT2232H, JTAG 3V3/1V8, UART		
Size: A3	Date: 2026-10-21	**Rev: 0.4**
KiCad E.D.A. 10.0.7		Id: 4/6

Gowin SPI flash (MSPI) — W25Q32JVZPIQ (replaces P25Q32U)



WP# and HOLD#/RESET# with 4.7k pull-ups (Tang Nano 9K 3674). EP not connected.

microSD (TF) in SPI mode



10k pull-ups on CMD and DAT0, 0402ESDA-05N ESD and R74 0R on CLK (Tang Nano 9K 3674).

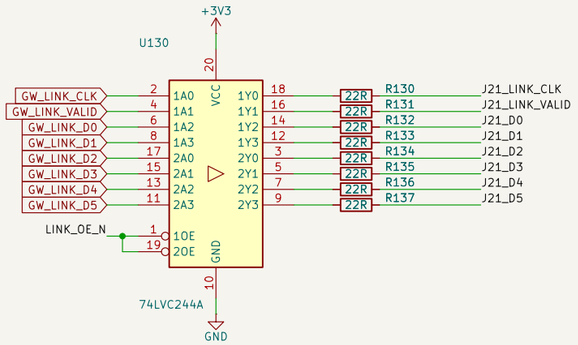
Preserved blocks: Tang Nano 9K 3674 (schematic + iBOM); FT2232H: FTDI DS v2.10 fig. 6.1
Pre-release (README rev 0.4): allocation and new parts are Proposed; open items in README section

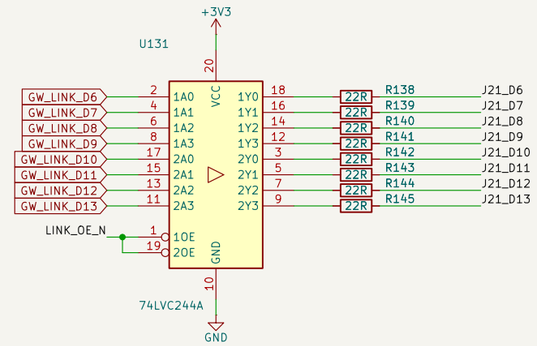
Sheet: /flash_microsd/
File: flash_microsd.kicad_sch

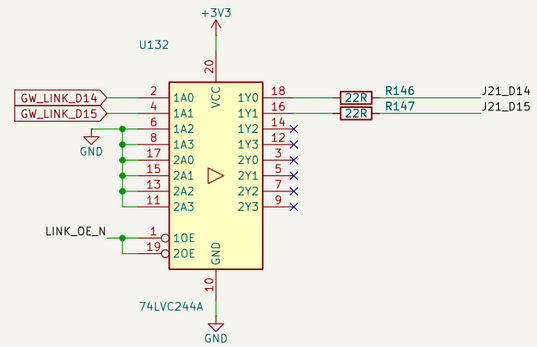
Title: SPI flash W25Q32JV and microSD

Size: A4	Date: 2026-10-21	**Rev: 0.4**
KiCad E.D.A. 10.0.7		Id: 5/6

Link outputs: 74LVC244A (loff) + 22R series



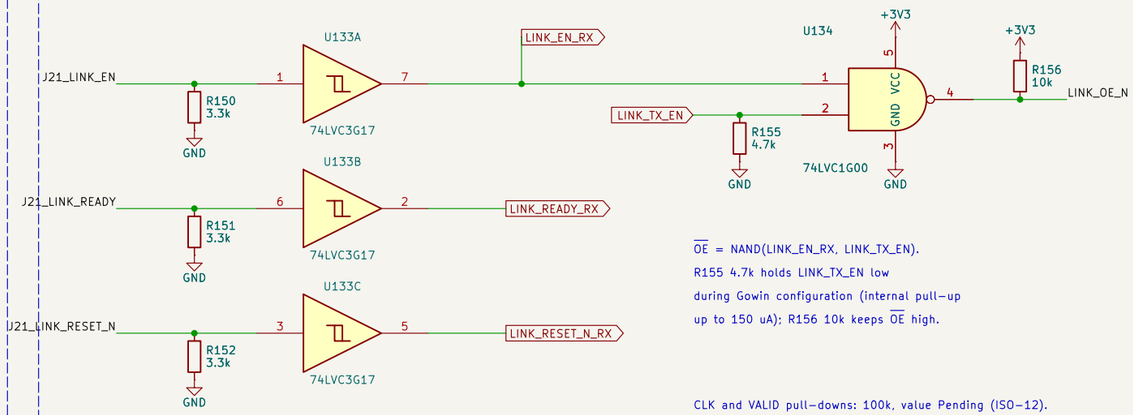




Outputs high-impedance (loff) with the KR260 off:

$\overline{OE}$ active only with KR260 LINK_EN and Gowin LINK_TX_EN (ISO-15).

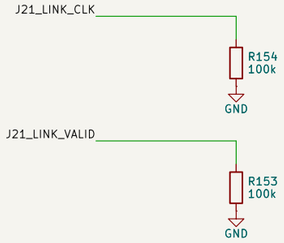
KR260 inputs (pull-down + 74LVC3G17 Schmitt, loff) and driver enable (ISO-15)



$\overline{OE}$ = NAND(LINK_EN_RX, LINK_TX_EN).

R155 4.7k holds LINK_TX_EN low during Gowin configuration (internal pull-up up to 150 uA); R156 10k keeps $\overline{OE}$ high.

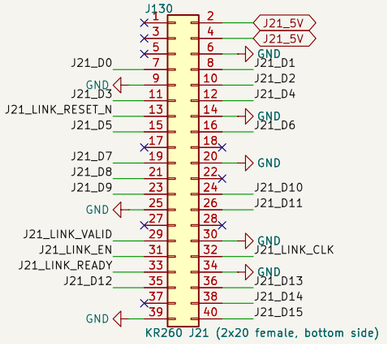
CLK and VALID pull-downs: 100k, value Pending (ISO-12).



Pull-down 3.3k: R <= VIL / K26 I_pullup (ISO-08).

74LVC3G17: Schmitt inputs with loff.

KR260 J21 (40 pins)



KR260 J21 (2x20 female, bottom side)

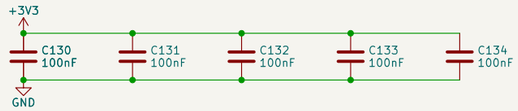
J21 -> K26 map: docs/pinout-j21.csv

(checked against KR260 sch. 038-05101-01 A02).

Pins 2 and 4 (KR260 5V) feed J21_5V (power sheet, OR-ed with USB-C).

Pins 1 and 17 (KR260 3V3): not connected.

+3V3 decoupling (U130-U134)



Preserved blocks: Tang Nano 9K 3674 (schematic + iBOM); FT2232H: FTDI DS v2.10 fig. 6.1

Pre-release (README rev 0.4): allocation and new parts are Proposed; open items in README section

Sheet: /j21_interface/

File: j21_interface.kicad_sch

Title: KR260 J21 interface: loff buffers, LINK_EN

Size: A3 | Date: 2026-10-21 | **Rev: 0.4**

KiCad E.D.A. 10.0.7 | Id: 6/6